

GlucoNova: A Hybrid Model for Non-Invasive Blood Glucose Monitoring Using Near-Infrared Photoplethysmography

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Abstract—Diabetes mellitus is a growing global health crisis requiring frequent and accurate blood glucose (BG) monitoring. Traditional invasive methods, while effective, present challenges of pain, high cost, and poor patient compliance. This study presents a novel non-invasive BG monitoring system leveraging a low-cost photoplethysmogram (PPG) sensor. The core innovation of this work is a hybrid prediction framework that combines a global polynomial regression model with an optional personalized model. This approach effectively addresses the issue of inter-subject physiological variability, which is a major limitation of many non-invasive techniques. The system's comprehensive pipeline includes a MATLAB-based signal processing stage and an automated data synchronization architecture that transmits real-time patient metrics to a secure, cloud-based platform for remote monitoring. Performance evaluation demonstrated the superior accuracy of the hybrid model yielding an outstanding RMSE of 0.602 mmol/L. This marked improvement confirms that our innovative hybrid methodology offers a precise and reliable solution for non-invasive BG monitoring, representing a significant step toward creating an accurate, affordable, and user-friendly system for personal diabetes management [1].

Index Terms—Non-invasive, blood glucose, photoplethysmography (PPG) signal, near-infrared (NIR), diabetes mellitus.

I. Introduction

Diabetes mellitus (DM) is a global chronic metabolic disorder that occurs when the insulin produced by the pancreas is not enough (Type 1 Diabetes) or the insulin produced by the body is not effectively utilized (Type 2 Diabetes). At present, there is no effective permanent cure for this disease but regular blood glucose monitoring is needed to reduce the complications of diabetes. The survey of World Diabetes Federation show that, this disease affects over 536 million adults worldwide in 2021 and this figure is expected to rise to 783 million by 2045 [2]. Uncontrolled diabetes can lead to severe complications such as cardiovascular disease, kidney failure, neuropathy and retinopathy [3]. Frequent monitoring of blood glucose is thus crucial for reducing the risk of complications. [4]

Traditional glucose monitoring techniques generally rely on invasive methods such as finger-prick tests and continuous glucose monitoring systems (CGMS), which involve piercing

the skin to obtain blood samples. Although clinically effective, these approaches are associated with significant drawbacks including pain, risk of infection, high cost and poor patient compliance, especially for individuals requiring multiple tests per day [5]. Therefore, the demand for a painless, affordable and reliable non-invasive glucose monitoring system has grown significantly in recent years.

Numerous non-invasive methods have been explored to address these challenges, including carbon nanotubes [6], polymeric materials [7], fluorescent [8], Raman spectroscopy [9], microwave [10] and nanobiosensors [8] techniques. Despite their potential, wearable healthcare is limited by their high cost, complexity, poor accuracy, and sensitivity to environmental and physiological conditions. Also, these models often fail to account for individual physiological characteristics, such as variations in bone and tissue thickness or fat composition, which can lead to reduced prediction accuracy [4], [11]. Photoplethysmography (PPG) has emerged as a promising non-invasive technique that captures volumetric changes in blood using light absorption properties of biological tissues [12]. Existing studies, while progressive, have not yet fully resolved key challenges in non-invasive BG monitoring. Therefore, it is crucial that future work investigates more effective signal processing techniques and advanced algorithms to improve predictive performance. [4] Numerous non-invasive methods have been explored to address these challenges, including carbon nanotubes [6], polymeric materials [7], fluorescent [8], Raman spectroscopy [9], microwave [10] and nanobiosensors [8] techniques. Despite their potential, wearable healthcare is limited by their high cost, complexity, poor accuracy, and sensitivity to environmental and physiological conditions. Also, these models often fail to account for individual physiological characteristics, such as variations in bone and tissue thickness or fat composition, which can lead to reduced prediction accuracy [4], [11]. Photoplethysmography (PPG) has emerged as a promising non-invasive technique that captures volumetric changes in blood using light absorption properties of biological tissues [12]. Existing studies, while progressive, have not yet fully resolved key challenges in non-invasive BG monitoring. Therefore, it is crucial that future work investigates more ef-

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fective signal processing techniques and advanced algorithms to improve predictive performance.

In this study, we propose a novel, low-cost, non-invasive glucose monitoring technique using the MAX30102 optical sensor operating in near infrared (NIR) wavelength which introduces a unique hybrid prediction framework that leverages a polynomial regression model to combine two distinct data sources: a global population-based model and a personalized, patient-specific model. Our approach uses a biasing method to intelligently blend predictions from both models, thereby achieving a balance between population-level trends and individual physiological characteristics. The proposed method eliminates the need for expensive spectroscopy or chemical sensors, offering an alternative that is simple, portable, and suitable for real-time applications. Data processing and glucose estimation are performed using MATLAB, and the results are visualized and transmitted to a cloud server for record-keeping and analysis. The experimental results demonstrate that the proposed hybrid model provides excellent predictive performance across a wide range of blood glucose concentrations, effectively leveraging both global and personalized data.

II. the Non-Invasive BG Monitoring System

A. Effect of Glucose on Absorption and Scattering in NIR

The MAX30102 sensor consists of a light source to illuminate the tissue and a photodiode to detect the reflected light. The amount of light absorbed varies with the fluctuations of blood volume [13]. When the light incident upon the fingertips, scattering and absorption occurs. As a result, the intensity of the light decreases [14], [15]. Glucose molecules have relatively low absorption in the NIR region. So, the reason for changing light intensity with the influence of glucose concentration is due to the refractive index difference between the extracellular fluid and the cell membrane [14]. This changes the path of light in the human tissue.

B. Optimal Wavelength Selection

The selection of an appropriate wavelength in the NIR spectrum plays a vital role in non-invasive glucose monitoring for effective tissue penetration. Biological tissues are composed of various organic molecules, including glucose, water, and hemoglobin, each possessing unique absorption characteristics within the NIR range [16]. Among the commonly explored wavelengths, the wavelengths from 810 to 850 nm, especially 880 nm, work well because they can specifically target the chemical bonds in glucose [3]. The 880 nm wavelength has been selected for its superior tissue penetration compared to visible light, allowing it to access blood vessels in the underlying tissue, particularly when the sensor is placed on anatomical sites such as the wrist or earlobe. Furthermore, the absorption of 880 nm light by water is minimal, which contributes to reduced signal attenuation and an improved signal-to-noise (SNR) ratio in the reflected light [3].

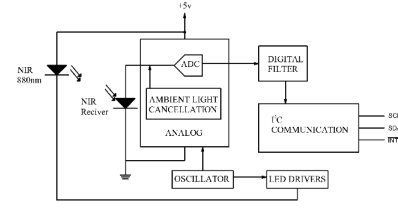


Fig. 1. Workflow of MAX30102 optical sensor.

TABLE I
KEY OPERATING PARAMETERS OF THE UTILIZED LED SENSING MODULE

Parameter	Value	Description
LED Brightness	31	Approx. 6.25 mA current
LED Mode	IR 880 nm	Infrared wavelength
Sample Averaging	8 samples per reading	Improves signal stability
Sampling Rate	100 Hz	100 samples per second
Pulse Width	411 μ s	Duration of LED pulse
ADC Range	4096	12-bit resolution

According to the Beer–Lambert law, the absorbance A is directly proportional to the concentration C of the absorbing species, as expressed in (1):

$$A = \varepsilon \cdot C \cdot l \quad (1)$$

In equation (1), ε represents the absorption coefficient, C signifies the concentration of glucose in the blood, and l is the optical path length within the tissue. To effectively extract glucose-related features from the reflected NIR signal, a multivariate regression technique was employed. This technique is capable of capturing the non-linear and multivariate dependencies between spectral features and glucose concentration, improving prediction accuracy.

C. Sensor Design

The MAX30102 sensor module, as shown in Fig. 1, serves as a key component of our non-invasive blood glucose monitoring system. This highly integrated pulse oximeter and heart rate sensor is specifically chosen for its high sensitivity, low power consumption, and robust performance in capturing PPG signals. The sensor operates by emitting light in the infrared spectrum and then measuring the light absorbed by the blood vessels. The MAX30102 module communicates with a microcontroller (Arduino Uno) using the standard I2C communication protocol. This configuration ensures that the digital data from the sensor is transmitted accurately and with minimal noise to the processing unit, which is crucial for the subsequent signal analysis and blood glucose estimation.

D. Sensor Configuration and Signal Acquisition

The MAX30102 sensor was interfaced using the MAX30105 Arduino library and configured with the following parameters to optimize signal quality, as described in Table I.

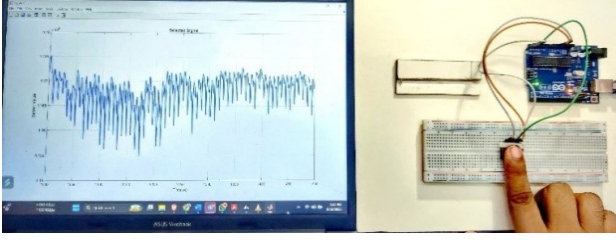


Fig. 2. Workflow of the overall system architecture.

III. System Architecture

The overall system architecture, as illustrated in Fig. 2, consists of three main components: an optical sensor, a microcontroller (Arduino Uno), and a personal computer for data processing and analysis. The optical sensor, placed under the fingertip, serves as the primary data acquisition unit. It captures the PPG signal by using a transmissive light path. Light from the sensor's LEDs passes through the finger and is detected by a photodiode. Pulsatile changes in blood volume cause variations in the detected light intensity, which are then converted into a measurable electrical signal.

The analog signal from the optical sensor is sent to a microcontroller. Its main function is to digitize the raw data using an Analog-to-Digital Conversion (ADC) process. The microcontroller is programmed to sample the PPG signal at a specific frequency, ensuring the data's accuracy and reliability for further analysis.

Finally, the digitized data is transmitted from the microcontroller to a Personal Computer via a serial port connection. The PC, running a powerful software environment like MATLAB, serves as the data processing and analysis hub. In MATLAB, the raw PPG signal undergoes various signal processing techniques, including filtering, feature extraction, and algorithm application, to ultimately estimate blood glucose levels. This setup provides a robust and flexible framework for real-time data collection and advanced analysis

IV. Feature Extraction and Training

A. Dataset Description

In this study, we collected a custom dataset from a total of 11 participants to train and validate our model. The cohort included 6 diabetic and 5 non-diabetic individuals, comprising 8 males and 3 females, with a mean age of 37.45 ± 1.3 years. Six of the participants were recruited from the RUET Medical Center. The remaining five participants were also measured with a standard invasive method (Accu-Chek Active blood glucose meter). All participants were informed about the study objectives and provided their consent prior to data acquisition.

Physiological signals were recorded from all participants under similar, controlled conditions while they were in a seated, resting position. The data was collected using a PPG sensor module based on the MAX30102 sensor. For each participant, a minimum 60-second PPG signal was recorded. To ensure data quality, we visually inspected each raw signal and selected the most stable and noise-free segments for better

Algorithm 1 Framework for Our Non-Invasive Blood Glucose Monitoring System

```

1: Start:
2: Initialize baseline:  $IR_0 \leftarrow \text{mean}(IR_{\text{raw}}[0 : N])$ 
3: Preprocess signal:
4:    $IR_{\text{sig}} \leftarrow IR_{\text{raw}} - \text{median}(IR_{\text{raw}})$ 
5:   Apply Butterworth low-pass filter:  $IR_{\text{filt}} \leftarrow \text{filt}(IR_{\text{sig}})$ 
6:   Normalize amplitude:  $IR_{\text{norm}} \leftarrow IR_{\text{filt}} / \max(|IR_{\text{filt}}|)$ 
7:   Select stable window ( $W$ ) from  $IR_{\text{norm}}$ 
8:   Extract segment:  $x \leftarrow IR_{\text{norm}}[W]$ ,  $t \leftarrow \text{time}[W]$ 
9:   Detect peaks:  $\text{peaks} \leftarrow \text{Peaks}(x)$ 
10:  Compute BPM:  $R \leftarrow \text{diff}(t_{\text{peaks}})$ ;  $BPM \leftarrow 60 / \text{mean}(R)$ 
11:  Detect troughs:  $\text{troughs} \leftarrow \text{Troughs}(x)$ 
12:  Calculate AC/DC ratio:
13:     $AC = \text{mean}(x_{\text{peaks}}) - \text{mean}(x_{\text{troughs}})$ 
14:     $DC = \text{mean}(x_{\text{troughs}})$ 
15:     $R = AC / DC$ 
16:  Estimate  $SpO_2$ :  $SpO_2 = 110 - 25 \cdot R$ 
17:  Predict glucose:
18:     $Y_g = f_{\text{global}}(BPM, SpO_2)$ 
19:     $Y_p = f_{\text{personal}}(BPM, SpO_2)$  (if available)
20:  Hybrid blending:
21:    if calibration data exists then
22:      optimize ( $m$ ) via least squares
23:    else
24:      use default ( $m$ )
25:    end if
26:     $Y_{\text{pred}} = m \cdot Y_g + (1 - m) \cdot Y_p$ 
27:  Store features: BPM,  $SpO_2$ ,  $R$ ,  $Y_g$ ,  $Y_p$ ,  $Y_{\text{pred}}$ ,  $m$ 
28:  if export enabled then
29:    Send JSON payload to cloud endpoint
30:  end if
31: End: Return  $Y_{\text{pred}}$ , feature set, blending weight ( $m$ ), and export status

```

analysis. The diabetic status of the participants was determined by obtaining their blood glucose levels using the traditional invasive method at the RUET Medical Center. This data served as the ground truth for validating the performance of our model.

algorithm algorithmic

B. Data Preprocessing

The raw PPG signal acquired through the NIR sensor is often contaminated by various sources of noise, including motion artifacts, muscle contractions, respiratory activity, and environmental interferences. To minimize such disturbances, subjects were instructed to remain seated and maintain a steady posture during data acquisition. However, despite these precautions, the acquired signals typically contain significant irregularities that must be addressed prior to meaningful analysis.

The overall data preprocessing procedure is shown in Fig. 3. At first, a segment of the PPG signal exhibiting minimal noise

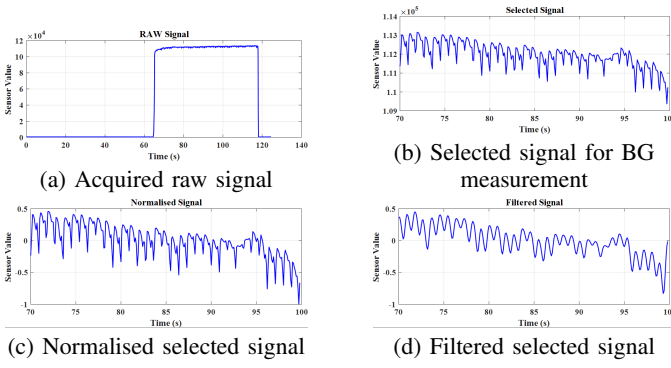


Fig. 3. Overall data preprocessing representation. (a) Acquired raw signal, (b) selected signal for BG measurement, (c) normalised selected signal, and (d) filtered selected signal.

and clear periodicity is manually selected for further analysis. To suppress high-frequency noise and enhance signal quality, a fourth-order Butterworth digital low-pass filter is applied with a normalized cut-off frequency of 0.4. The normalized cut-off frequency of 0.4 Hz was selected based on empirical testing across multiple recordings, and it was observed that this value consistently produced the best results in terms of noise suppression and waveform preservation.

C. BPM and SpO₂ Calculation

Heart rate is estimated from the filtered PPG signal using a peak detection approach. After normalizing the signal, the PPG `peakfind` function is utilized to identify the locations of local maxima in the PPG waveform, each representing a heartbeat, as shown in Fig. 4. The heart rate is then computed based on the inverse of the mean inter-peak interval, scaled to a per-minute value using equation (2), where Δt represents the time difference between successive peaks:

$$\text{BPM} = \frac{60}{\text{mean}(\Delta t)} \quad (2)$$

SpO₂ is estimated by analyzing the amplitude characteristics of the filtered PPG signal. The AC component of the PPG signal is approximated by the average peak amplitude, while the DC component is represented by the average trough amplitude. The ratio R of these two components is computed in equation (3):

$$R = \frac{AC}{|DC|} \quad (3)$$

Using this ratio, SpO₂ is calculated through a widely adopted empirical formula based on optical absorption characteristics [17], as shown in equation (4):

$$\text{SpO}_2(\%) = 110 - 25R \quad (4)$$

D. Feature Extraction and Selection

Accurate estimation of blood glucose levels requires going beyond basic physiological parameters such as BPM and SpO₂, as these alone may not sufficiently capture the complex,

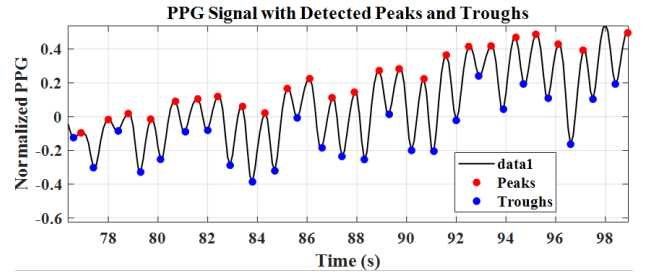


Fig. 4. PPG peak detection and waveform analysis for BPM and SpO₂ estimation.

TABLE II
REGRESSION COEFFICIENTS AND STATISTICAL SIGNIFICANCE FOR GLUCOSE PREDICTION MODEL

Term	Estimate	Standard Error (SE)	t-Statistic	p-Value
Intercept	909.51	103.40	8.7956	4.9518×10^{-5}
x_1	0.037643	0.010485	3.5900	0.008857
x_2	-19.095	2.2011	-8.6750	5.4150×10^{-5}
x_3	0.10051	0.011704	8.5876	5.7816×10^{-5}

non-linear relationships inherent in physiological processes. To address this limitation, polynomial regression was employed to incorporate higher-order interactions and enhance predictive performance. Polynomial features were engineered from the sensor outputs, specifically by including the square of SpO₂ as an additional feature to model non-linear dependencies.

The selected features used in the regression model are:

$$x_1 = \text{BPM}, \quad x_2 = \text{SpO}_2, \quad x_2^2 = (\text{SpO}_2)^2$$

The regression model is expressed in equation (5):

$$y = a + b \cdot x_1 + c \cdot x_2 + d \cdot x_2^2 \quad (5)$$

Here, y represents the predicted blood glucose level, and a, b, c, d are the model coefficients. The estimated coefficients and their associated statistics for the polynomial regression model are summarized in Table II. The determination of these coefficients was done using multiple linear regression via MATLAB's `fitlm()` function, which optimizes the model parameters by minimizing the Mean Squared Error (MSE) between predicted and actual glucose values.

The final fitted polynomial regression model derived from empirical data is expressed in equation (6):

$$\text{Glucose Level (mmol/L)} = 909.51 + 0.038 \cdot \text{BPM} - 19.09 \cdot \text{SpO}_2 + 0.10 \cdot (\text{SpO}_2)^2 \quad (6)$$

E. Personalized Dataset Construction

To enhance the accuracy of non-invasive blood glucose level estimation, a personalized dataset construction approach is adopted. From the PPG signal, two key physiological parameters are extracted—BPM and SpO₂. These features serve as independent variables in the modeling process. Using

MATLAB, a personalized polynomial regression model is developed specifically for the patient. This regression equation is calibrated based on the patient's physiological characteristics and historical blood glucose measurements.

The output of this model is referred to as the personalized predicted glucose level, denoted in equation (7):

$$Y_p = f_{\text{personal}}(\text{BPM}, \text{SpO}_2) \quad (7)$$

F. Biasing Method: Hybrid Prediction Framework

To ensure both population-level trends and individual-specific dynamics are incorporated, a biasing method is used that blends predictions from two models—a global model and a personalized model. A structured Microsoft Excel dataset is used, containing the following variables for each data point: Y_{real} (actual glucose values measured invasively), Y_g (predicted glucose values from the global regression model), and Y_p (predicted glucose values from the personalized regression model). The final glucose level prediction Y_{pred} is computed using a weighted combination of the global and personalized predictions, as shown in equation (8):

$$Y_{\text{pred}} = m \cdot Y_g + (1 - m) \cdot Y_p \quad (8)$$

Here, $m \in [0, 1]$ is a biasing coefficient that determines the influence of the global versus personalized models. The value of m is optimized by minimizing the Mean Squared Error (MSE) between the predicted and actual glucose levels across all observations, as shown in equation (9):

$$\text{MSE} = \min_m \sum_{i=1}^n [Y_{\text{real},i} - (m \cdot Y_{g,i} + (1 - m) \cdot Y_{p,i})]^2 \quad (9)$$

This optimization ensures that the hybrid model adapts to the individual while still leveraging global trends. The final model yields more accurate and personalized glucose predictions, particularly effective in real-time continuous monitoring scenarios.

G. Overall System Pipeline

The pipeline of our non-invasive blood glucose (BG) monitoring system is illustrated in Fig. 5, which operates through a structured, multi-stage workflow.

It begins with the acquisition of a raw PPG signal from an NIR sensor. This signal is processed in MATLAB, where a noise-free segment is selected and filtered. The core contribution of this work lies in its unique hybrid prediction framework. This model intelligently blends predictions from a population-based model with individual patient data, significantly enhancing prediction accuracy by accounting for unique physiological variations. The processed data, including estimated BG, BPM, and SpO_2 , is automatically transmitted to a Google Apps Script-based cloud platform for real-time monitoring and record-keeping.

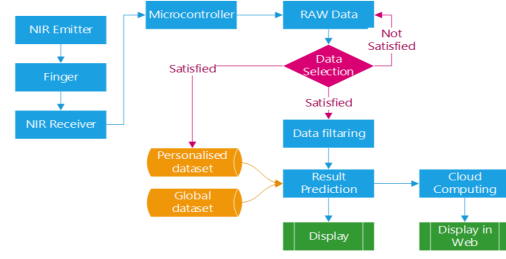


Fig. 5. End-to-end pipeline for our non-invasive blood glucose monitoring system.



Fig. 6. Pipeline for Cloud-Based Data Synchronization for Clinical Monitoring.

TABLE III
PERFORMANCE METRICS OF THE GLOBAL REGRESSION MODEL FOR GLUCOSE PREDICTION

RMSE (mmol/L)	R ²	Adjusted R ²	MAE (mmol/L)
0.696	0.935	0.907	0.567

H. Data Export and Synchronization Architecture

To ensure real-time data access and security, we developed a fully automated data pipeline using MATLAB and Google Apps Script, as depicted in Fig. 6. After local signal processing, key patient metrics are transmitted via an HTTP POST request to a dedicated web app. The web app securely stores data in a Google Sheet for continuous monitoring and analysis.

I. Web Interface

The web interface serves as a user-friendly dashboard for real-time patient data visualization, presented in Fig. ???. The interface is designed to be intuitive and informative, allowing patients and healthcare providers to monitor key metrics without specialized software. The complete interface is hosted at:

<https://sites.google.com/view/glucose-level-detection/home>

V. Result and Discussion

The predictive capability of our system was evaluated using two distinct models: a global polynomial regression model and an advanced hybrid prediction framework.

A. Global Model Performance

The global model, trained on a population-based dataset of 11 observations, demonstrated robust performance. The model's effectiveness is highlighted by the key performance metrics listed in Table III.

Fig. 7 visualizes the trends between actual and predicted glucose levels for our global model. The low Root Mean Squared Error (RMSE) signifies a small average difference between the predicted and actual values, indicating strong model performance.

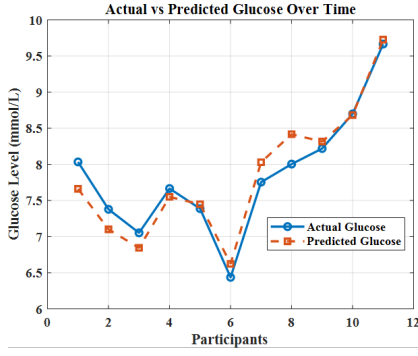


Fig. 7. Actual vs predicted glucose level graph for global model.

TABLE IV
PERFORMANCE METRICS OF THE HYBRID REGRESSION MODEL FOR
GLUCOSE PREDICTION

RMSE (mmol/L)	R ²	Adjusted R ²	MAE (mmol/L)
0.602	0.964	0.921	0.535

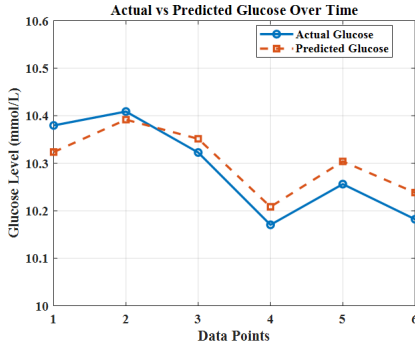


Fig. 8. Actual vs predicted glucose level graph for hybrid model.

B. Hybrid Model Performance

A significant advancement of this study is the development of a hybrid prediction framework that is trained on 6 data points from a single participant and combines the global model with a personalized model. This approach addresses the limitation of population-based models by accounting for individual physiological differences. The effectiveness of this method is clearly shown in the results for a single participant, as detailed in Table IV.

The actual and predicted glucose levels for our hybrid model are visualized in Fig. 8. By integrating the personalized model with the global model using an optimized biasing method, the final hybrid model reached an outstanding RMSE of just 0.602 mmol/L. This remarkable reduction in error demonstrates that combining global and personalized data significantly enhances the predictive accuracy of the system, offering a more reliable solution for individual users.

C. Comparison with Existing Methods

The model's performance was validated through a comprehensive comparative analysis against existing methodologies

TABLE V
COMPARATIVE ANALYSIS WITH EXISTING STUDIES

Studies	Method	RMSE	R ²	Limitations
Guo et al. [5]	PPG, HRV	1.380	0.69	Need extensive computation
Zihan Nie et al. [18]	NIR, Feature extraction, PPG	1.253	0.60	Dataset primarily from non-diabetic individuals
Abdolrazzaghi et al. [19]	Microwave based	0.519	0.98	Based on simulated data only
Our model	hybrid PPG, NIR, Feature extraction	0.602	0.96	—

in the field of non-invasive glucose monitoring, as depicted in Table V.

D. Comparison with Existing Methods

Our hybrid model's performance is highly competitive and, in many cases, superior to other methods, especially when considering the low-cost nature of our sensor design. Many existing methods either rely on more complex spectroscopy techniques or require large, diverse datasets. Our method provides a viable, low-cost alternative with exceptional accuracy, making it highly suitable for widespread use in home healthcare settings. The predictive performance of our hybrid model, with an RMSE of 0.602 mmol/L and R² value of 0.96, is highly competitive with existing non-invasive methods. Our approach not only provides accurate predictions but also addresses limitations found in other studies. Compared to the work of Guo et al. [5], our hybrid model offers a more efficient approach that avoids high computational complexity. It also improves upon the model by Zihan Nie et al. [18] by using a more diverse, real-world dataset rather than one from a limited population. Furthermore, our method is validated with real human data, setting it apart from studies based solely on simulations, such as the one by Abdolrazzaghi et al. [19]. In conclusion, our hybrid model offers a practical and accurate solution for real-time glucose monitoring by overcoming the common deficiencies of previous approaches.

VI. Conclusion

We proposed a novel, low-cost, non-invasive blood glucose monitoring technique to address the limitations of existing methods. Our approach uses a hybrid prediction framework that combines a population-based global model with a personalized, patient-specific model. This innovation leverages a low-cost MAX30102 optical sensor to capture PPG signals in the NIR region, effectively accounting for individual physiological variations. Our system achieved an outstanding RMSE of 0.602 mmol/L and a high R² value of 0.96, confirming its superior predictive accuracy compared to a purely global model. We also implemented an automated data pipeline that securely exports and synchronizes real-time patient data to a cloud platform for continuous monitoring. Despite these promising results, the current system has limitations including a small participant group, a non-wearable form factor, and

a need for manual calibration. Future work will focus on acquiring a larger, more diverse dataset, optimizing the device for portability, and conducting real-world testing to validate its robustness. In summary, this study presents a novel and effective methodology that offers a reliable, precise, and affordable alternative for diabetes management.

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